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A Closer Look at Lithium Ion – The World's Favorite Battery

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Lithium-ion batteries power nearly everything with a charge that isn't plugged into a wall socket. Your phones, tablets, laptops, Tesla electric cars and more are all powered by this one form of battery. How did we get here? And how long are we staying with this tried and true battery format?

Nickel Cadmium (NiCad)

Back in the technological dark ages of the 1980s, trinkets with the novelty of a rechargeable battery generally came with nickel cadmium batteries. This early form of rechargeable battery was burdensome to recharge because it required a full discharge beforehand. This was cumbersome in practice and increased recharge time.

NiCad also had the unfortunate ability of overcharging. Since these batteries “consist of a pressure vessel (<http://bit.ly/1g1pyZC>) that is supposed to contain any generation of oxygen and hydrogen gases until they can recombine back to water,” there was a potential that a cheap charging unit wouldn't be sophisticated enough to recognize the delicate balance needed to maintain cell integrity while charging. This led to a many NiCad bat-

teries to quickly lose the ability to hold a charge.

Nickel Metal Hydride (NiMH)

Entering consumer grade in 1989 (<http://bit.ly/1OLnyAe>), NiMH became the successor to NiCad along with a more expensive competitor the Li-ion in many gadgets. NiMH quickly took hold in high-drain devices like digital cameras and became a staple in early hybrid vehicles. Benefits of this form of battery include reliability not seen in NiCad and a lower cost than the Lithium Ion battery, which at the time made a huge difference.



Lithium Ion (Li-ion)

Lithium-ion batteries first became commercially viable back in 1991. This was long before the idea of carrying one in your pocket at all times became commonplace. Significantly more expensive than previous forms of rechargeable batteries, the lithium-ion offered a bevy of unique (<http://bit.ly/1DnzUh6>) features. It offered a higher energy output than nickel metal hydride while resolving many of the issues that plagued nickel cadmium batteries as well.

What many people might not know is the broad range of types of batteries that fall under the lithium-ion battery umbrella. Here are a few of the most common in use:

- **Lithium Cobalt Oxide** – a popular type for laptops, cell phones and digital cameras. “The drawback of Li-cobalt is a relatively short life span,” says (<http://bit.ly/1SCVvsU>) Battery University. Users will normally see degradation in capacity after about 500 recharge cycles. This is why you’ll often note how quick your cellphone battery drains about two years into owning it. This form is also susceptible to heat, which you might have experienced when attempting to use an iPhone on an exceedingly hot day. The iPhone is smart enough to know when to power down due to excessive heat to protect battery integrity and minimize the risk of damaging the battery that could lead to it exploding into flames in some instances.
- **Lithium Nickel Manganese Cobalt Oxide** – A type of Frankenstein battery that combines two cathode materials. Nickel and manganese benefit each other by combining strengths. Nickel adds a high energy output while normally offering poor stability when holding a charge, but manganese is used to create stability in the battery while it would normally give poor output. The two together creates a high output with better stability, essentially benefitting each other through a symbiotic relationship. Each of the material’s strength overrides the other’s weaknesses.

- **Lithium Manganese Oxide** – This form of lithium-ion battery is also known as 18650 and will start making more news in the near future as this is the type that will be used in the Tesla Powerwall (<http://bit.ly/1edlHa9>). The battery's spiral structure allows for three-dimensional flow of ions which is said to improve ion flow and would result in a lower internal resistance. This improves current handling and should help provide a high output when multiple appliances are pulling from the stored energy.

Stu Statman at Greentechmedia explains (<http://bit.ly/1KMDwJJ>) why Tesla may have selected this form of battery for its home-powering battery:

"Many large 18650 batteries can be cycled 2,500 times or so before they lose much of their storage efficacy. Prismatics [meaning three-dimensional cube cell configuration where other cells operate in a two-dimensional sheet configuration] can retain a high percentage of their storage capacity, however, after being cycled 7,500 to 10,000 times, making them better suited for applications where their power will be regularly drawn all the way down and charged back up."



Takeaway

With the creation of Tesla's Gigafactory in Nevada, the market will soon be flooded with lithium-ion batteries. The expressed intention of this massive increase in production is to drive down cost and increase the availability of this form of battery.

It seems our future will be powered on lithium-ion, at least for the foreseeable future. There may be emerging battery tech on the horizon, but for now lithium-ion batteries are the go-to source for storing energy off the grid.

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